2.69 2D Matrices: Diagonalisation

Worked Solutions

1 a)

For $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$,

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5).$$

So $\lambda_1 = 2$ and $\lambda_2 = 5$.

$$\lambda_1 = 2: \quad \mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}, \quad 2x + y = 0, \quad y = -2x, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

$$\lambda_2 = 5: \quad \mathbf{A} - 5\mathbf{I} = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}, \quad -x + y = 0, \quad y = x, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

$$\therefore \mathbf{P} = \begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix}.$$

1 b)

$$\mathbf{P}^{-1} = \frac{1}{3}\begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}.$$

$$\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \frac{1}{3}\begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 6 & 0 \\ 0 & 15 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}.$$

$$\therefore \mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}.$$

2 a)

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (7-\lambda)(6-\lambda) - 6 = \lambda^2 - 13\lambda + 36 = (\lambda - 4)(\lambda - 9).$$

$$\therefore \lambda = 4 \text{ or } \lambda = 9.$$

When $\lambda = 4$, $\mathbf{A} - 4\mathbf{I} = \begin{pmatrix} 3 & 2 \\ 3 & 2 \end{pmatrix}$, so $3x + 2y = 0$ and $\mathbf{v}_1 = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$. When $\lambda = 9$, $\mathbf{A} - 9\mathbf{I} = \begin{pmatrix} -2 & 2 \\ 3 & -3 \end{pmatrix}$, so $x = y$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

$$\therefore \mathbf{P} = \begin{pmatrix} 2 & 1 \\ -3 & 1 \end{pmatrix}, \quad \mathbf{P}^{-1} = \frac{1}{5}\begin{pmatrix} 1 & -1 \\ 3 & 2 \end{pmatrix}.$$

$$\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \frac{1}{5}\begin{pmatrix} 1 & -1 \\ 3 & 2 \end{pmatrix}\begin{pmatrix} 7 & 2 \\ 3 & 6 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ -3 & 1 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 20 & 0 \\ 0 & 45 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}.$$

$$\therefore \mathbf{P} = \begin{pmatrix} 2 & 1 \\ -3 & 1 \end{pmatrix} \text{ and } \mathbf{D} = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}.$$

2 b)

$$\det(\mathbf{A}-\lambda\mathbf{I}) = (20-\lambda)(17-\lambda) - 180 = \lambda^2 - 37\lambda + 160 = (\lambda-5)(\lambda-32).$$

$$\therefore \lambda = 5 \text{ or } \lambda = 32.$$

When $\lambda_1 = 4$

$$\mathbf{A} - 5\mathbf{I} = \begin{pmatrix} 15 & 12 \\ 15 & 12 \end{pmatrix}, \quad 15x + 12y = 0, \quad 5x = -4y, \quad \mathbf{v}_1 = \begin{pmatrix} 4 \\ -5 \end{pmatrix}.$$

When $\lambda_2 = 32$, $\mathbf{A} - 32\mathbf{I} = \begin{pmatrix} -12 & 12 \\ 15 & -15 \end{pmatrix}$, so $x = y$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

$$\therefore \mathbf{P} = \begin{pmatrix} 4 & 1 \\ -5 & 1 \end{pmatrix}, \quad \mathbf{P}^{-1} = \frac{1}{9}\begin{pmatrix} 1 & -1 \\ 5 & 4 \end{pmatrix}.$$

$$\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \frac{1}{9}\begin{pmatrix} 1 & -1 \\ 5 & 4 \end{pmatrix}\begin{pmatrix} 20 & 12 \\ 15 & 17 \end{pmatrix}\begin{pmatrix} 4 & 1 \\ -5 & 1 \end{pmatrix} = \frac{1}{9}\begin{pmatrix} 45 & 0 \\ 0 & 288 \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 0 & 32 \end{pmatrix}.$$

$$\therefore \mathbf{P} = \begin{pmatrix} 4 & 1 \\ -5 & 1 \end{pmatrix} \text{ and } \mathbf{D} = \begin{pmatrix} 5 & 0 \\ 0 & 32 \end{pmatrix}.$$

2 c)

$$\det(\mathbf{A}-\lambda\mathbf{I}) = (12-\lambda)(-9-\lambda) + 98 = \lambda^2 - 3\lambda - 10 = (\lambda+2)(\lambda-5).$$

$$\therefore \lambda = -2 \text{ or } \lambda = 5.$$

When $\lambda_1 = -2$, $\mathbf{A} + 2\mathbf{I} = \begin{pmatrix} 14 & 7 \\ -14 & -7 \end{pmatrix}$, so $y = -2x$ and $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$. When $\lambda_2 = 5$, $\mathbf{A} - 5\mathbf{I} = \begin{pmatrix} 7 & 7 \\ -14 & -14 \end{pmatrix}$, so $x = y$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

$$\therefore \mathbf{P} = \begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix}, \quad \mathbf{P}^{-1} = \begin{pmatrix} -1 & -1 \\ 2 & 1 \end{pmatrix}.$$

$$\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} -1 & -1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 12 & 7 \\ -14 & -9 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ 0 & 5 \end{pmatrix}.$$

$$\therefore \mathbf{P} = \begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix} \text{ and } \mathbf{D} = \begin{pmatrix} -2 & 0 \\ 0 & 5 \end{pmatrix}.$$

3 a)

$$\det(\mathbf{A}-\lambda\mathbf{I}) = (5-\lambda)^2, \quad \lambda = 5 \text{ or } \lambda = 5.$$

When $\lambda = 5$, $\mathbf{A} - 5\mathbf{I} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, so $y = 0$ and $\mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

3 b)

A matrix is only diagonalisable if it has enough linearly independent eigenvectors and in this case there is only one independent eigenvector.

4 a)

He incorrectly constructs matrices $\mathbf{P}$ and $\mathbf{D}$; he should have used

$$\begin{pmatrix} 7 & 0 \\ 0 & 16 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 16 & 0 \\ 0 & 7 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}.$$

When computing $\mathbf{P}^{-1}$ he did not divide by the determinant.

4 b)

$$\mathbf{P} = \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{P}^{-1} = \frac{1}{3} \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}.$$

$$\mathbf{PDP}^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 7 & 0 \\ 0 & 16 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 39 & 18 \\ 9 & 30 \end{pmatrix} = \begin{pmatrix} 13 & 6 \\ 3 & 10 \end{pmatrix}.$$

5 a)

$$\begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix}^8 = \begin{pmatrix} 5^8 & 0 \\ 0 & 4^8 \end{pmatrix} = \begin{pmatrix} 390625 & 0 \\ 0 & 65536 \end{pmatrix}.$$

5 b)

$$\begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}^{10} = \begin{pmatrix} 1^{10} & 0 \\ 0 & (-2)^{10} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1024 \end{pmatrix}.$$

5 c)

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}^k = \begin{pmatrix} a^k & 0 \\ 0 & b^k \end{pmatrix}.$$

6)

$$\begin{aligned} \mathbf{M}^n &= (\mathbf{PDP}^{-1})^n \\ &= (\mathbf{PDP}^{-1})(\mathbf{PDP}^{-1})(\mathbf{PDP}^{-1}) \cdots (\mathbf{PDP}^{-1}) \\ &= \mathbf{PD}(\mathbf{P}^{-1}\mathbf{P})\mathbf{D}(\mathbf{P}^{-1}\mathbf{P})\mathbf{D} \cdots (\mathbf{P}^{-1}\mathbf{P})\mathbf{DP}^{-1} \\ &= \mathbf{PDDD} \cdots \mathbf{DDP}^{-1} = \mathbf{PD}^n\mathbf{P}^{-1} \quad \text{as req.} \end{aligned}$$

7 a)

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (3 - \lambda)(2 - \lambda), \quad \lambda = 2 \text{ or } \lambda = 3.$$

When $\lambda_1 = 2$, $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, so $y = -x$ and $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. When $\lambda_2 = 3$, $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix}$, so $y = 0$ and x can take any value, giving $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

$$\therefore \lambda_1 = 2,\ \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ and } \lambda_2 = 3,\ \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

7 b)

$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$.

7 c)

$$\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP} \Rightarrow \mathbf{PD} = \mathbf{AP} \Rightarrow \mathbf{PDP}^{-1} = \mathbf{A} \therefore \mathbf{A}^4 = \mathbf{PD}^4\mathbf{P}^{-1}.$$

$$\mathbf{P}^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}.$$

$$\mathbf{A}^4 = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}^4 \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 16 & 0 \\ 0 & 81 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 81 & 65 \\ 0 & 16 \end{pmatrix}.$$

8 a)

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (-4 - \lambda)(1 - \lambda), \quad \lambda = -4 \text{ or } \lambda = 1.$$

8 b)

When $\lambda_1 = -4$, $\mathbf{A} + 4\mathbf{I} = \begin{pmatrix} 0 & 0 \\ -15 & 5 \end{pmatrix}$, so $y = 3x$ and $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$. When $\lambda_2 = 1$, $\mathbf{A} - \mathbf{I} = \begin{pmatrix} -5 & 0 \\ -15 & 0 \end{pmatrix}$, so $x = 0$ and y can take any value, giving $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

8 c)

$\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} -4 & 0 \\ 0 & 1 \end{pmatrix}$.

8 d)

$$\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP} \Rightarrow \mathbf{PDP}^{-1} = \mathbf{A} \therefore \mathbf{A}^4 = \mathbf{PD}^4\mathbf{P}^{-1}.$$

$$\mathbf{P}^{-1} = \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}.$$

$$\mathbf{A}^{50} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} -4 & 0 \\ 0 & 1 \end{pmatrix}^{50} \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 2^{100} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} 2^{100} & 0 \\ 3 \times 2^{100} - 3 & 1 \end{pmatrix}.$$

9 a)

$$\mathbf{P} = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{pmatrix}, \quad \mathbf{P}^{-1} = \frac{1}{2}\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

$$\mathbf{M}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 4^{-n} \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 & 1-4^{-n} \\ 0 & 2\times 4^{-n} \end{pmatrix} = \begin{pmatrix} 1 & \frac{1}{2}(1-4^{-n}) \\ 0 & 4^{-n} \end{pmatrix}.$$

9 b)

$$\mathbf{L} = \lim_{n\to\infty}\begin{pmatrix} 1 & \frac{1}{2}(1-4^{-n}) \\ 0 & 4^{-n} \end{pmatrix} = \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 0 \end{pmatrix}.$$

10 a)

$$\det(\mathbf{A}-\lambda\mathbf{I}) = \left(\frac{2}{3}-\lambda\right)\left(\frac{5}{6}-\lambda\right) - \frac{1}{18} = \lambda^2 - \frac{3}{2}\lambda + \frac{1}{2} = \frac{1}{2}(2\lambda-1)(\lambda-1).$$

$$\therefore \lambda_1 = \frac{1}{2}, \quad \lambda_2 = 1.$$

When $\lambda_1 = \frac{1}{2}$, $\mathbf{A} - \frac{1}{2}\mathbf{I} = \begin{pmatrix} \frac{1}{6} & \frac{1}{3} \\ \frac{1}{6} & \frac{1}{3} \end{pmatrix}$, so $y = -\frac{1}{2}x$ and $\mathbf{v}_1 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$. When $\lambda_2 = 1$, $\mathbf{A} - \mathbf{I} = \begin{pmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{1}{6} & -\frac{1}{6} \end{pmatrix}$, so $y = x$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

$$\mathbf{P} = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{P}^{-1} = \frac{1}{3}\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}.$$

$$\mathbf{M}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 2^{-n} & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2^{1-n}+1 & 1-2^{1-n} \\ 1-2^{-n} & 2+2^{-n} \end{pmatrix}.$$

10 b)

$$\mathbf{L} = \lim_{n\to\infty}\frac{1}{3}\begin{pmatrix} 2^{1-n}+1 & 1-2^{1-n} \\ 1-2^{-n} & 2+2^{-n} \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

11)

$$\mathbf{P} = \begin{pmatrix} 3 & 1 \\ -2 & 0 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}, \quad \mathbf{P}^{-1} = \frac{1}{2}\begin{pmatrix} 0 & -1 \\ 2 & 3 \end{pmatrix}.$$

$$\begin{aligned} \mathbf{A}^n &= \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1} \\ &= \frac{1}{2}\begin{pmatrix} 3 & 1 \\ -2 & 0 \end{pmatrix}\begin{pmatrix} 2^n & 0 \\ 0 & 4^n \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 2 & 3 \end{pmatrix} \\ &= \frac{1}{2}\begin{pmatrix} 3 & 1 \\ -2 & 0 \end{pmatrix}\begin{pmatrix} 0 & -2^n \\ 2\times 4^n & 3\times 4^n \end{pmatrix} \\ &= \frac{1}{2}\begin{pmatrix} 2\times 4^n & 3\times 4^n - 3\times 2^n \\ 0 & 2\times 2^n \end{pmatrix} \end{aligned}$$

$$= \begin{pmatrix} 4^n & \frac{3}{2}(4^n - 2^n) \\ 0 & 2^n \end{pmatrix} \quad \text{as req.}$$

12)

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (2-\lambda)(3-\lambda) - 6 = \lambda^2 - 5\lambda = \lambda(\lambda - 5).$$
$$\therefore \lambda_1 = 0, \quad \lambda_2 = 5.$$

When $\lambda_1 = 0$, $\mathbf{Av} = \mathbf{0}$ gives $x + 3y = 0$, so $\mathbf{v}_1 = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$. When $\lambda_2 = 5$, $\mathbf{A} - 5\mathbf{I} = \begin{pmatrix} -3 & 6 \\ 1 & -2 \end{pmatrix}$, so $x = 2y$ and $\mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

$$\mathbf{P} = \begin{pmatrix} -3 & 2 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 0 & 0 \\ 0 & 5 \end{pmatrix}, \quad \mathbf{P}^{-1} = -\frac{1}{5}\begin{pmatrix} 1 & -2 \\ -1 & -3 \end{pmatrix}.$$

$$\begin{aligned} \mathbf{A}^n &= \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1} \\ &= -\frac{1}{5}\begin{pmatrix} -3 & 2 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & 5^n \end{pmatrix}\begin{pmatrix} 1 & -2 \\ -1 & -3 \end{pmatrix} \\ &= -\frac{1}{5}\begin{pmatrix} -3 & 2 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ -5^n & -3\times 5^n \end{pmatrix} \\ &= \frac{1}{5}\begin{pmatrix} 2\times 5^n & 6\times 5^n \\ 5^n & 3\times 5^n \end{pmatrix} \\ &= 5^{n-1}\begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix} \quad \text{as req.} \end{aligned}$$

13)

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (p-\lambda)(q-\lambda) - pq = \lambda^2 - \lambda(p+q) = \lambda(\lambda - (p+q)).$$
$$\therefore \lambda = 0 \text{ or } \lambda = p + q.$$

When $\lambda_1 = -k$ then $\mathbf{A} + 0\mathbf{I} = \begin{pmatrix} p & q \\ p & q \end{pmatrix}$

$$\mathbf{A}\begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{0} \Rightarrow px + qy = 0, \quad qy = -px, \quad \mathbf{v}_1 = \begin{pmatrix} q \\ -p \end{pmatrix}.$$

When $\lambda_2 = p + q$, $\mathbf{A} - (p+q)\mathbf{I} = \begin{pmatrix} -q & q \\ p & -p \end{pmatrix}$, so $y = x$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

$$\mathbf{P} = \begin{pmatrix} q & 1 \\ -p & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 0 & 0 \\ 0 & p+q \end{pmatrix}, \quad \mathbf{P}^{-1} = \frac{1}{p-q}\begin{pmatrix} 1 & -1 \\ p & q \end{pmatrix}.$$

$$\mathbf{A}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1} = \frac{1}{p+q}\begin{pmatrix} q & 1 \\ -p & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & (p+q)^n \end{pmatrix}\begin{pmatrix} 1 & -1 \\ p & q \end{pmatrix}$$

$$= \frac{1}{p+q}\begin{pmatrix} p(p+q)^n & q(p+q)^n \\ p(p+q)^n & q(p+q)^n \end{pmatrix} = (p+q)^{n-1}\begin{pmatrix} p & q \\ p & q \end{pmatrix} = (p+q)^{n-1}\mathbf{A} \quad \text{as req.}$$

14)

$$\det(\mathbf{A}-\lambda\mathbf{I}) = (1-\lambda)(-1-\lambda) - (k+1)(k-1) = \lambda^2 - k^2 = (\lambda+k)(\lambda-k).$$

$$\therefore \lambda = -k \text{ or } \lambda = k.$$

When $\lambda_1 = -k$, $\mathbf{A} + k\mathbf{I} = \begin{pmatrix} 1+k & k+1 \\ k-1 & -1+k \end{pmatrix}$, so $x = -y$ and $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. When $\lambda_2 = k$, $\mathbf{A} - k\mathbf{I} = \begin{pmatrix} 1-k & k+1 \\ k-1 & -1-k \end{pmatrix}$, so $y = \frac{k-1}{k+1}x$ and $\mathbf{v}_2 = \begin{pmatrix} k+1 \\ k-1 \end{pmatrix}$.

$$\mathbf{P} = \begin{pmatrix} 1 & k+1 \\ -1 & k-1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} -k & 0 \\ 0 & k \end{pmatrix}, \quad \mathbf{P}^{-1} = \frac{1}{2k}\begin{pmatrix} k-1 & -(k+1) \\ 1 & 1 \end{pmatrix}.$$

$$\mathbf{A}^{2n+1} = \mathbf{P}\mathbf{D}^{2n+1}\mathbf{P}^{-1} = \frac{1}{2k}\begin{pmatrix} 1 & k+1 \\ -1 & k-1 \end{pmatrix}\begin{pmatrix} -k^{2n+1} & 0 \\ 0 & k^{2n+1} \end{pmatrix}\begin{pmatrix} k-1 & -(k+1) \\ 1 & 1 \end{pmatrix}$$

$$= \frac{1}{2}k^{2n}\begin{pmatrix} 1 & k+1 \\ -1 & k-1 \end{pmatrix}\begin{pmatrix} -k+1 & k+1 \\ 1 & 1 \end{pmatrix} = k^{2n}\begin{pmatrix} 1 & k+1 \\ k-1 & -1 \end{pmatrix} = k^{2n}\mathbf{A}^n \quad \text{as req.}$$